Majid Ghasemi

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Research interests

Social Reinforcement Learning, Multi-agent Reinforcement Learning, Machine Ethics, AI Alignment, and AI Ethics

Education

University of Waterloo Waterloo, ON, Canada

Ph.D. in Electrical & Computer Engineering Sep 2025 – Present
Supervised by Mark Crowley
Research area: *Ethical decision making in multi-agent systems and how social learning can be a way to achieve it*

Wilfrid Laurier University (WLU) Waterloo, ON, Canada
MASc in Applied Computing Sep 2023 – May 2025
Supervised by Dariush Ebrahimi. *GPA: A+.*
Thesis title: *Reinforcement Learning for Scheduling and Routing: Electric Vehicles Charging and Patrol Scheduling*

Imam Khomeini Intl. University (IKIU) Qazvin, Iran
B.Sc. in Software Engineering Sep 2017 – March 2022
Supervised by Mahdi Bahaghighat. *GPA: A-.*

Publications

Citation count prediction based on Google Scholar profiles and Clarivate's journal citation reports
Mahdi Bahaghighat, Leila Akbari, *Majid Ghasemi*, and Qin Xin.
Information Research journal

Toward Virtuous Reinforcement Learning: A Critique and Roadmap
Majid Ghasemi, and Mark Crowley.
Machine Ethics: from formal methods to emergent machine ethics workshop at the AAAI conference

Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response
Majid Ghasemi, and Dariush Ebrahimi.
IECON 2025 – 51st Annual Conference of the IEEE Industrial Electronics Society.

Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety

Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, and Dariush Ebrahimi.
2025 IEEE 34th International Symposium on Industrial Electronics (ISIE).

Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging
Majid Ghasemi, Fadi Alzhouri, and Dariush Ebrahimi.
2025 21st International Conference on the Design of Reliable Communication Networks (DRCN).

Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges
Majid Ghasemi, Amirhossein Mousavi, and Dariush Ebrahimi.
Preprint on Arxiv.

Introduction to Reinforcement Learning
Majid Ghasemi, and Dariush Ebrahimi.
Preprint on Arxiv.

Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm
Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, *Majid Ghasemi*, Fadi Alzhouri, and Dariush Ebrahimi.
2024 International Conference on the Dynamics of Information Systems.

A high-accuracy phishing website detection method based on machine learning
Mahdi Bahaghighat, *Majid Ghasemi*, and Figen Ozen.
Journal of Information Security and Applications 77.

Honors and scholarships

Winner of the Academic Excellence Gold Medal (WLU)	2025
GSRDA Award (University of Waterloo)	2026
Int'l Doctoral Student Award (University of Waterloo)	2025 – 2029
Graduate Scholarship (WLU)	2023 – 2025
Graduate Fellowship (WLU)	2023 – 2025
Differential Tuition Fee Waiver (WLU)	2023 – 2025
Excellence Students Tuition Fee Waiver (IKIU)	2017–2022

Research experience

UWECEML/CompThink lab

Supervisor: Prof. Mark Crowley (University of Waterloo) Sep 2025 – Present
Developing socially informed and ethically grounded approaches to multi-agent reinforcement learning.

Edge Computing lab
Supervisor: Dr. Dariush Ebrahimi (WLU) Sep 2023 – May 2025
Conducted research on reinforcement learning for intelligent transportation systems, focusing on EV fleet management and routing optimization.

AIST lab
Supervisor: Dr. Mahdi Bahaghighat (IKIU) Sep Jan. 2021 – March 2022
Researched on phishing website detection with machine learning.

Teaching experience

Teaching Assistant, IKIU, WLU, and UW 2023-Present

Courses including AI, Data Mining, Algorithm Design and Analysis, and Discrete Mathematics

Industry experience

Kisoji Biotechnology Inc., Machine Learning Department Waterloo, ON, Canada
AI Research Intern May 2026 – Present
Currently building the Expressibility Engine — a 3-module system for antibody expression optimization: domain-adaptive prediction with Bayesian project embeddings, RL-guided experimental design (GRPO + DPP diversity), and sparse graph-based causal driver discovery. Reduces wet-lab rounds via active learning while providing interpretable guidance for antibody engineers.

Service to the Community

(R) Stands for Reviewer, (PC) stands for Program Committee

(PC) AAAI/ACM Conference on AI, Ethics, and Philosophy (AIES) 2026
(R) Reinforcement Learning in Big Worlds at Reinforcement Learning Conference (RLC) 2026
(R) PhilML Workshop at ICML 2026
(R) Pluralistic Alignment Workshop at ICML 2026
(R) Post-AGI Science and Society (P-AGI) Workshop at ICLR 2026
(R) IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)
(R) ACM Transactions on Intelligent Systems and Technology

Grants

HU University of Applied Sciences Utrecht Aug. 2022 – Aug. 2023

Recipient of a grant to conduct research on Woman, Life, Freedom movement in Iran with computational social science and machine learning approaches.